Solution Requirements

Date	15 March 2024
Team ID	UNM-293
Project Name	AI Comic Story Creator using Gemini Models
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

· Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).
· Non-Functional Requirements (NFR) specify criteria that judge operation of a system (How system should be).

Step 1: Functional Requirements (FR)

S.No	Category	Description	Priority
1	Authentication	User authentication via session - no login needed for MVP, API key secured via .env & Streamlit secrets	Medium
2	Authorization	Single user role for MVP - User can generate comics; Admin (Dhashini) manages API keys & deployment	Low
3	External Interfaces	Google Gemini API (gemini-1.5-flash for story, Imagen for images), Streamlit Cloud hosting	High
4	Transactions Processing	Story generation transaction: Input → Gemini → Story → Prompt builder → Image generation → Comic assembly	High
5	Reporting	Export comic as PDF/images, Display comic history, Usage stats (no. of comics generated)	Medium
6	Business Rules	Max 6 panels per comic, Story length 300-500 words, Image prompts auto-enhanced with style tags	High
7	Compliance	Google AI API terms compliance, No storage of API keys in code, Secure handling via .env	High
8	Other	Custom genre/character input, Preview before download, Sample comics gallery	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Description	Target Metric
1	Performance & Speed	Response time for story <2 sec, Image gen <5 sec, Page load <2 sec	Page loads <2s, API avg 1.42s
2	Scalability	Support up to 50 concurrent users, Handle 100 comics/day, Auto scaling via cloud	50 users, 45 req/sec
3	Security & Data Privacy	API key in .env, No user data stored, HTTPS, Secure secrets management	AES-256 env, HTTPS
4	Reliability & Availability	99% uptime, Graceful failure handling for Gemini quota, Retry logic	99% uptime/month
5	Usability & Accessibility	Simple UI, Responsive design, WCAG 2.1 AA, Loading indicators, Tutorial	Complies WCAG 2.1
6	Other (Maintainability)	Modular code, PEP8, Git version control, Docker ready, Easy API swap	Maintainable